

Lecture 3 : Series

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1. Infinite series

Let (u_n) be a sequence. Then the summation $u_1 + u_2 + u_3 + \dots = \sum_{n=1}^{\infty} u_n$ or $\sum u_n$ is said to be a infinite series (or a series) generated by the sequence (u_n) . u_n is said to be the n^{th} **term** of the series.

Let the sequence (s_n) defined by

$$s_1 = u_1, s_2 = u_1 + u_2, s_3 = u_1 + u_2 + u_3, \dots$$

The elements of the sequence (s_n) are called the **partial sum** of the series $\sum u_n$.

The sequence (s_n) is called the sequence of partial sums of the series $\sum u_n$.

The infinite series $\sum u_n$ is said to be convergent or divergent according as the sequence of partial sum (s_n) is convergent or divergent.

Convergent series : The infinite series $\sum u_n$ is said to be **convergent** if the sequence of its partial sum (s_n) is convergent. If sequence (s_n) converges to a real number s then s is called the sum of the series and we write $\sum u_n = s$.

Divergent series : The infinite series $\sum u_n$ is said to be **divergent** to ∞ or $-\infty$ if the sequence (s_n) is divergent to ∞ or $-\infty$ and we write $\sum u_n = \infty$ or $\sum u_n = -\infty$.

Oscillatory series : The infinite series $\sum u_n$ is said to be **oscillating finitely** or **infinitely** if the sequence of its partial sum (s_n) is oscillating finitely or infinitely.

Examples

1. Let us consider the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

Comparing it with the series $\sum_{n=1}^{\infty} u_n$, we have $u_n = \frac{1}{n(n+1)}$.

Let $s_n = u_1 + u_2 + u_3 + \dots$

Then,

$$\begin{aligned} s_n &= \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}, \end{aligned}$$

and $\lim s_n = 1$. Hence the series $\sum u_n$ is convergent and the sum of the series is 1.

2. Let us consider the series $1 + 2 + 3 + \dots$

Let $s_n = 1 + 2 + 3 + \dots + n$. Then $s_n = \frac{n(n+1)}{2}$ and $\lim s_n = \infty$. Hence, the series is divergent.

2. Cauchy's principle of convergence

A necessary and sufficient condition for the convergence of a series $\sum u_n$ is that corresponding to a preassigned positive number ϵ , there exists a natural number m such that $|u_{n+1} + u_{n+2} + \dots + u_{n+p}| < \epsilon$ for all $n \geq m$ and for every natural number p .

Proof: Let $\sum u_n$ is convergent. Then, the sequence (s_n) is convergent. Therefore, by Cauchy's principle of convergence for the sequence, corresponding to a preassigned positive number ϵ , there exists a natural number m , such that

$|s_{n+p} - s_n| < \epsilon$ for all $n \geq m$ and for every natural number p .

Or, $|u_{n+1} + u_{n+2} + \dots + u_{n+p}| < \epsilon$ for all $n \geq m$ and for every natural number p .

Conversely, let us assume that for a preassigned positive number ϵ , there exists a natural number m , such that

$|u_{n+1} + u_{n+2} + \dots + u_{n+p}| < \epsilon$ for all $n \geq m$ and for every natural number p .

Then, $|s_{n+p} - s_n| < \epsilon$ for all $n \geq m$ and for every natural number p .

This implies that the sequence (s_n) is convergent by Cauchy's principle of convergence.

Therefore, $\sum u_n$ is convergent.

This completes the proof.

Theorem 1 A necessary condition for the convergence of a series $\sum u_n$ is $\lim u_n = 0$.

Proof: Let $\sum u_n$ be convergent series. Then for a preassigned positive number ϵ , there exists a natural number m , such that

$|u_{n+1} + u_{n+2} + u_{n+3} + \dots + u_{n+p}| < \epsilon$ for all $n \geq m$ and for every natural number p .

Taking $p = 1$, $|u_{n+1}| < \epsilon$ for all $n \geq m$. This implies $\lim u_n = 0$.

Note : But above theorem not sufficiently holds.

For example, let us consider the series $\sum u_n$ where $u_n = \frac{1}{n}$.

Here $\lim u_n = 0$. But $\sum u_n$ is a divergent series.

Here, $|s_{n+p} - s_n| = \left| \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+p} \right|$.

If we take $p = n$, we have

$$\begin{aligned} |s_{n+p} - s_n| &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \\ &> \frac{1}{2n} + \frac{1}{2n} + \dots + \frac{1}{2n} = \frac{1}{2}. \end{aligned}$$

Therefore, $|s_{n+p} - s_n|$ cannot be made less than a chosen positive number $\epsilon < \frac{1}{2}$ for every natural number p . Hence series $\sum \frac{1}{n}$ is divergent.

3. Comparison test

Let $\sum u_n$ and $\sum v_n$ be two series of positive real numbers and there exists a natural number m such that $u_n \leq kv_n$ for all $n \geq m$, k being a fixed positive number. Then

1. $\sum u_n$ is convergent if $\sum v_n$ is convergent.
2. $\sum v_n$ is divergent if $\sum u_n$ is divergent.

Proof: Let $s_n = u_1 + u_2 + \dots + u_n$, $t_n = v_1 + v_2 + \dots + v_n$. Then

$$\begin{aligned} s_n - s_m &= u_{m+1} + u_{m+2} + \dots + u_n \\ &\leq k(v_{m+1} + v_{m+2} + \dots + v_n) \\ &= k(t_n - t_m) \end{aligned}$$

or, $s_n \leq kt_n + h$ where $h = s_m - kt_m$, a finite number.

1. Let $\sum v_n$ be convergent. Then, the sequence (t_n) is bounded. Let B be an upper bound. Then, $t_n < B$ for all $n \in \mathbb{N}$.
Therefore, $s_n < kB + h$ for all $n \geq m$.
This shows that the sequence (s_n) is bounded above. (s_n) being a monotone increasing sequence bounded above, is convergent. Therefore, $\sum u_n$ is convergent.
2. Let $\sum u_n$ is divergent. Then the sequence (s_n) is not bounded above. Since $s_n \leq kt_n + h$, the sequence (t_n) is not bounded above.
Therefore the series $\sum v_n$ is divergent.

Example : Test the convergence of the series $\sum 2^n$.

Solution : Let $\sum u_n$ be the given series. Then $u_n = 2^n$.

Let $v_n = \frac{1}{n}$. Then, $v_n < u_n$ for all $n \geq 1$. We proved earlier that series $\sum \frac{1}{n}$ is divergent. Therefore, from the comparison test, the series $\sum 2^n$ is divergent.

Theorem 2 The series $\sum \frac{1}{n^p}$, converges for $p > 1$ and diverges for $p \leq 1$

Proof:

Case 1. $p > 1$.

Let $\sum_{n=1}^{\infty} u_n$ be the given series. Then $u_n = \frac{1}{n^p}$.

Let $\sum v_n$ be obtained from $\sum u_n$ by grouping the terms as

$$1 + \left(\frac{1}{2^p} + \frac{1}{3^p} \right) + \left(\frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} \right) + \left(\frac{1}{8^p} + \dots + \frac{1}{15^p} \right) + \dots$$

Then

$$\begin{aligned} v_1 &= 1, \quad v_2 = \frac{1}{2^p} + \frac{1}{3^p} < \frac{1}{2^p} + \frac{1}{2^p} = \frac{1}{2^{p-1}}, \\ v_3 &= \frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} < \frac{4}{4^p} = \frac{1}{2^{2(p-1)}}, \\ v_4 &= \frac{1}{8^p} + \dots + \frac{1}{15^p} < \frac{8}{8^p} = \frac{1}{2^{3(p-1)}}, \\ \dots &\quad \dots \quad \dots \quad \dots \end{aligned}$$

Let $w_n = \left\{\frac{1}{2^{p-1}}\right\}^{n-1}$. Then, $v_n < w_n$ for all $n \geq 2$. But $\sum w_n$ is a geometric series of common ratio $\frac{1}{2^{p-1}}$.

Since $p > 1$, $0 < \frac{1}{2^{p-1}} < 1$ and hence $\sum w_n$ is convergent.

Therefore, $\sum v_n$ is convergent by Comparison test.

Since $\sum u_n$ is a series of positive terms and $\sum v_n$ is obtained from $\sum u_n$ by introduction of brackets and so, $\sum u_n$ is also convergent.

Case 2. $p = 1$.

In this case the series is

$$1 + \frac{1}{2} + \frac{1}{3} + \dots$$

$$\text{Here, } |s_{n+p} - s_n| = \left| \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+p} \right|.$$

If we take $p = n$, we have

$$\begin{aligned} |s_{n+p} - s_n| &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \\ &> \frac{1}{2n} + \frac{1}{2n} + \dots + \frac{1}{2n} = \frac{1}{2}. \end{aligned}$$

Therefore, $|s_{n+p} - s_n|$ cannot be made less than a chosen positive number $\epsilon < \frac{1}{2}$ for every natural number p . Therefore series $\sum \frac{1}{n}$ is divergent.

Case 3. $0 < p < 1$.

Then,

$$\frac{1}{n^p} > \frac{1}{n} \text{ for all } n \geq 2.$$

But $\sum \frac{1}{n}$ is divergent. Therefore, series $\sum \frac{1}{n^p}$ is divergent by Comparison test.

Case 4. $p \leq 0$.

$\lim_{n \rightarrow \infty} \frac{1}{n^p} \neq 0$, Therefore, series $\sum \frac{1}{n^p}$ is not convergent.

This completes the proof.

Examples

1. Series $\sum \frac{1}{n^2}$. Here $p = 2 > 1$. Therefore, series $\sum \frac{1}{n^2}$ is convergent.
2. Series $\sum n$. Here $p = -1 < 1$. Therefore, series $\sum n$ is divergent.

4. D' Alembert's ratio test

Let $\sum u_n$ be a series of positive real numbers and let $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$. Then $\sum u_n$ is

1. convergent if $l < 1$,

2. divergent if $l > 1$,
3. test fail to give a decision when $l = 1$.

Proof:

Case 1. $l < 1$.

Let us choose a positive number ϵ such that $l + \epsilon < 1$.

Since $\lim \frac{u_{n+1}}{u_n} = l$, there exists a natural number m such that

$$l - \epsilon < \frac{u_{n+1}}{u_n} < l + \epsilon, \text{ for all } n \geq m.$$

Let $l + \epsilon = r$. Then $0 < r < 1$.

We have $\frac{u_{m+1}}{u_m} < r, \frac{u_{m+2}}{u_{m+1}} < r, \dots, \frac{u_n}{u_{n-1}} < r$ where $n > m$.

Consequently, $\frac{u_n}{u_m} < r^{n-m}$ for all $n > m$, or $u_n < \frac{u_m}{r^m} \cdot r^n$ for all $n > m$.

$\frac{u_m}{r^m}$ is a positive number and $\sum r^n$ is a geometric series of common ratio r where $0 < r < 1$ and therefore, $\sum r^n$ is convergent.

Therefore, $\sum u_n$ is convergent by Comparison test.

Case 2. $l > 1$.

Let us choose a positive number ϵ such that $l - \epsilon > 1$.

Since, $\lim \frac{u_{n+1}}{u_n} = l$, there exists a natural number k such that

$$l - \epsilon < \frac{u_{n+1}}{u_n} < l + \epsilon, \text{ for all } n \geq k.$$

Let $l - \epsilon = p$. Then $p > 1$.

We have $\frac{u_{k+1}}{u_k} > p, \frac{u_{k+2}}{u_{k+1}} > p, \dots, \frac{u_n}{u_{n-1}} > p$ where $n > k$.

Consequently, $\frac{u_n}{u_k} > p^{n-k}$ for all $n > k$. or $u_n > \frac{u_k}{p^k} \cdot p^n$ for all $n > k$.

$\frac{u_k}{p^k}$ is a positive number and $\sum p^n$ is a geometric series of common ratio $p > 1$ and therefore $\sum p^n$ is divergent.

Therefore, $\sum u_n$ is divergent by Comparison test.

Case 3. $l = 1$, the test fails to give a decision.

Counter example : Let $u_n = \frac{1}{n}$. Here, $\lim \frac{u_{n+1}}{u_n} = 1$, and series $\sum u_n$ is a divergent series.

Let $u_n = \frac{1}{n^2}$. Here, $\lim \frac{u_{n+1}}{u_n} = 1$, and series $\sum u_n$ is a convergent series.

Although, for both the series $\lim \frac{u_{n+1}}{u_n} = 1$, but one of the series is convergent and the other is a divergent.

Therefore, if $\lim \frac{u_{n+1}}{u_n} = 1$, nothing can be said about the convergence or divergence of the series $\sum u_n$.

Examples

1. Test the convergence of the series

$$1 + \frac{3}{2!} + \frac{5}{3!} + \frac{7}{4!} + \dots$$

Solution : Let $\sum u_n$ be the given series. Then, $u_n = \frac{2n-1}{n!}$. Then

$$\frac{u_{n+1}}{u_n} = \frac{2n+1}{(n+1)(2n-1)} \text{ and } \lim \frac{u_{n+1}}{u_n} = 0 < 1.$$

By D'Alembert's ratio test, $\sum u_n$ is convergent.

2. Test the convergence of the series

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots, \quad x > 0.$$

Solution : Let $\sum u_n$ be the given series. Since $x > 0$, $\sum u_n$ is a series of positive terms. Then, $u_n = \frac{x^n}{n}$. Then,

$$\frac{u_{n+1}}{u_n} = \frac{nx}{n+1} \text{ and } \lim \frac{u_{n+1}}{u_n} = x.$$

By D'Alembert's ratio test, $\sum u_n$ is convergent if $x < 1$ and $\sum u_n$ is divergent if $x > 1$. When $x = 1$, the series becomes $1 + \frac{1}{2} + \frac{1}{3} + \dots$, and this is divergent series.

Reference book : *Introduction to real analysis by S.K. Mapa.*